Predicting Movie Revenue From Movie Budget

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Summary

This study investigates how movie budget predicts domestic box office revenue using the Bechdel Test movie dataset from FiveThirtyEight. Exploratory anaylsis revealed right-skewed distributions for both budget and revenue, which were log-transformed to stabilize variance. A linear regression model and a K-nearest neighbours (KNN) regression model were fitted to predict log-transformed domestic revenue from log-transformed movie budget. Both models showed moderate predictive performance, with a root mean squared prediction error (RMSPE) of roughly 1.27 and 1.28 log units, respectively. The results suggest that while higher budgets are generally associated with higher revenue, budget alone cannot fully explain the variability in box office performance. These findings highlight the complexity of film revenue prediction and the importance of incorporating additional factors beyond production budget.

Introduction

The financial performance of a film is often assumed to be strongly related to its production budget. Higher budgets allow studios to invest in elements such as well-known actors, special effects, marketing campaigns, and larger production teams, which are often assumed to influence audience appeal and box office revenue (Basuroy et al. 2003). However, the extent to which movie budgets actually predict revenue remains uncertain, as many lower-budget films have seen significant success in the box office while some high-budget productions fail to recover the costs associated with making the movie (De Vany and Walls 1999).

Understanding the relationship between production budget and revenue can provide insight into whether increased financial investments reliably leads to higher financial returns.

Research Question

How does movie budget predict movie revenue?

The Data Set

In this analysis, we use FiveThirtyEight's Bechdel Test dataset, which contains descriptive information, production budgets and domestic box office revenue of 1,615 films released between 1990 and 2013 to answer our research question (FiveThirtyEight 2014).

Here, we take a look at the first couple rows of the raw data.

Table 1: First 6 rows of the raw data

year	imdb	title	test	clean_test	binary	budget	domgross	intgross	code	budget_2013	domgross_2013	intgross_2013	period.code	decade.code
2013	tt1711425	21 & Over	notalk	notalk	FAIL	1.30e+07	25682380	42195766	2013FAIL	13000000	25682380	42195766	1	1
2012	tt1343727	Dredd 3D	ok-disagree	ok	PASS	4.50e+07	13414714	40868994	2012PASS	45658735	13611086	41467257	1	1
2013	tt2024544	12 Years a Slave	notalk-disagree	notalk	FAIL	2.00e+07	53107035	158607035	2013FAIL	20000000	53107035	158607035	1	1
2013	tt1272878	2 Guns	notalk	notalk	FAIL	6.10e+07	75612460	132493015	2013FAIL	61000000	75612460	132493015	1	1
2013	tt0453562	42	men	men	FAIL	4.00e+07	95020213	95020213	2013FAIL	40000000	95020213	95020213	1	1
2013	tt1335975	47 Ronin	men	men	FAIL	2.25e+08	38362475	145803842	2013FAIL	225000000	38362475	145803842	1	1

Cleaning Data

Since our research question only requires studying 2 variables (movie budget - "budget", movie revenue - "domgross"), we will filter the data to only observe those 2 columns during our analysis.

Table 2: First 6 rows of the data

budget	domgross
1.30e+07	25682380
4.50e+07	13414714
2.00e+07	53107035
6.10e+07	75612460
4.00e+07	95020213
2.25e+08	38362475

Next, we will check our data for any null values and filter them out if there are any.

```
# of null values (before filtering): 17
```

```
# of null values (after filtering): 0
```

We will also check for any budget/revenue with a value of 0, since any movie with a $0 budget/revenue is most likely missing data. In checking, we found that before filtering, there was 0 movies with no budget and 1 movie with no revenue.

After seeing there is one entry with a non-zero budget but $0 revenue, we will first check to see what this value looks like. Since the value has a very high budget, we will say this entry is most likely having its appropriate budget missing and filter it out of the data set.

Table 3: Movies with a $0 revenue

budget	domgross
1.8e+07	0

EDA

Here, we will look at some general summary statistics of our data:

Table 4: Summary statistics of the cleaned movie dataset

X	budget	domgross
Min.	7000	828
1st Qu.	12000000	16320870
Median	30000000	42239614
Mean	45174408	69170974
3rd Qu.	60000000	93382814
Max.	425000000	760507625

Here, we will look at the general shape/values in our data:

Table 5: Structure of the cleaned movie dataset

budget	domgross
numeric	numeric

Next, we will use different plots to examine the skew/behaviour of budget and revenue in the data.

In Figure 1, we can see there is a heavy right skew in both variables, seeing as most of the movies are clustered at the bottom with only few movies pulling the tail towards higher values.

In Figure 2, we can once again observe how both variance are heavily right skewed, and like in the box plots above, we see only some movies with higher values which indicates we have some outliers in our data. In addition, after seeing both the box plot and histogram, we observe that the values have a very wide range and spread.

Next, we will look at the behaviour of movie budget with movie revenue on the untransformed data to help us gather an idea of how linear the relationship may be. From observing Figure 3, we can see that the data may have a positive linear relationship. The spread of the values are also very wide, making it difficult to conclude how strong this linear relationship may be.

Since just observing the scatter plot makes it difficult to conclude how linear our data is, we can observe the correlation between movie budget and revenue. We can see that we have a correlation of around 0.627, indicating that there may be a moderate positive relationship between our two variables. This correlation value allows us to further our analysis to see how well movie budget may predict movie revenue.

Previously, we observed a very large spread. To account for this, we will apply a variance stabilizing function (log) to our variables to help stabilize the wide range of values. We can see in Figure 4 that applying this function helps make the data less spread out and easier to observe for patterns.

Here is our data with the log-transformed budget and revenue added:

Table 6: First six rows with budget, domestic gross, and their log-transformed values

budget	domgross	log_budget	log_domgross
1.30e+07	25682380	16.38046	17.06132
4.50e+07	13414714	17.62217	16.41186
2.00e+07	53107035	16.81124	17.78782
6.10e+07	75612460	17.92638	18.14113
4.00e+07	95020213	17.50439	18.36960
2.25e+08	38362475	19.23161	17.46259

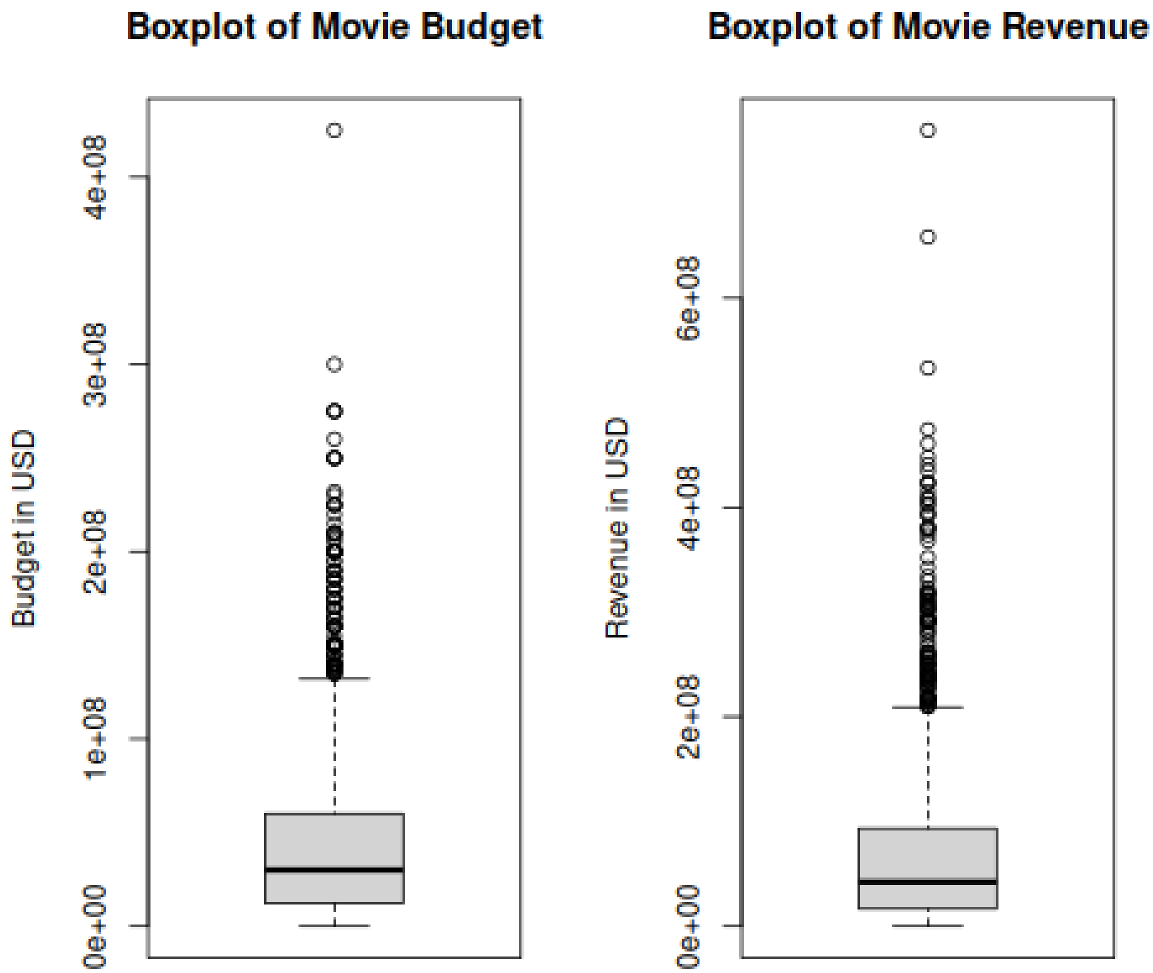

Figure 1: Boxplots of Movie Budget and Revenue

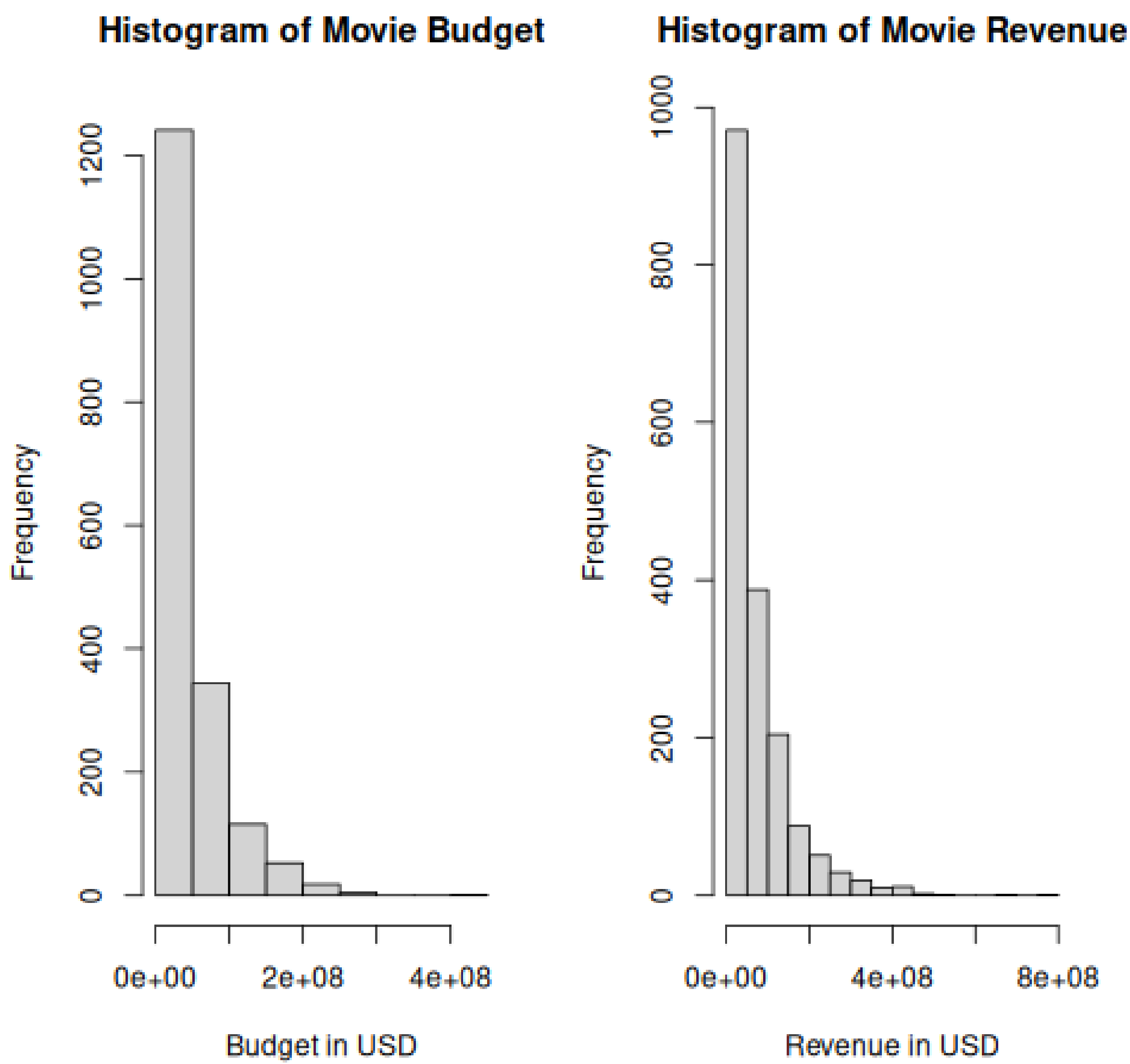

Figure 2: Histograms of Movie Budget and Revenue

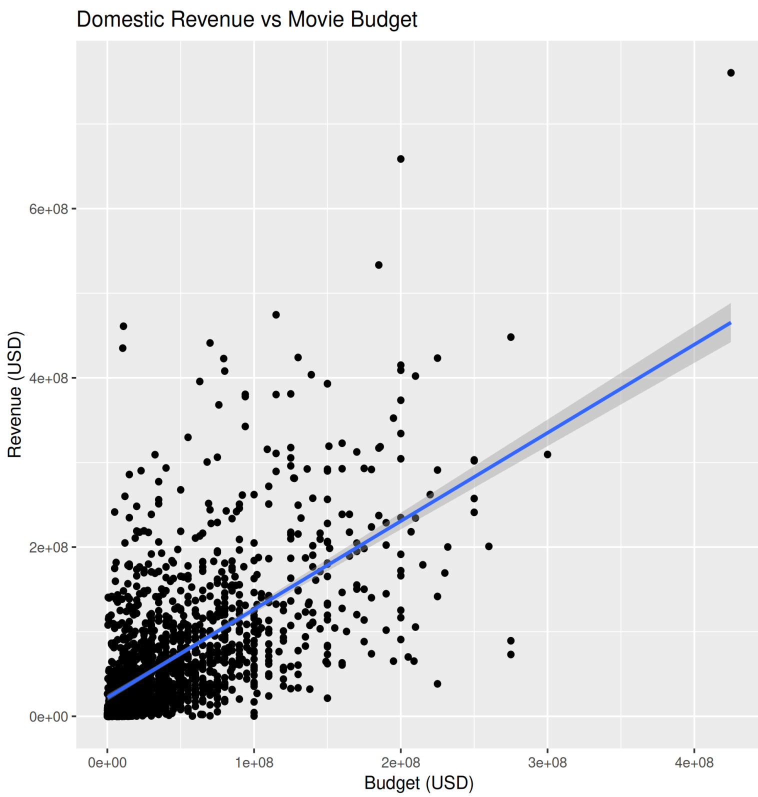

Figure 3: Scatterplot of Domestic Revenue vs Movie Budget

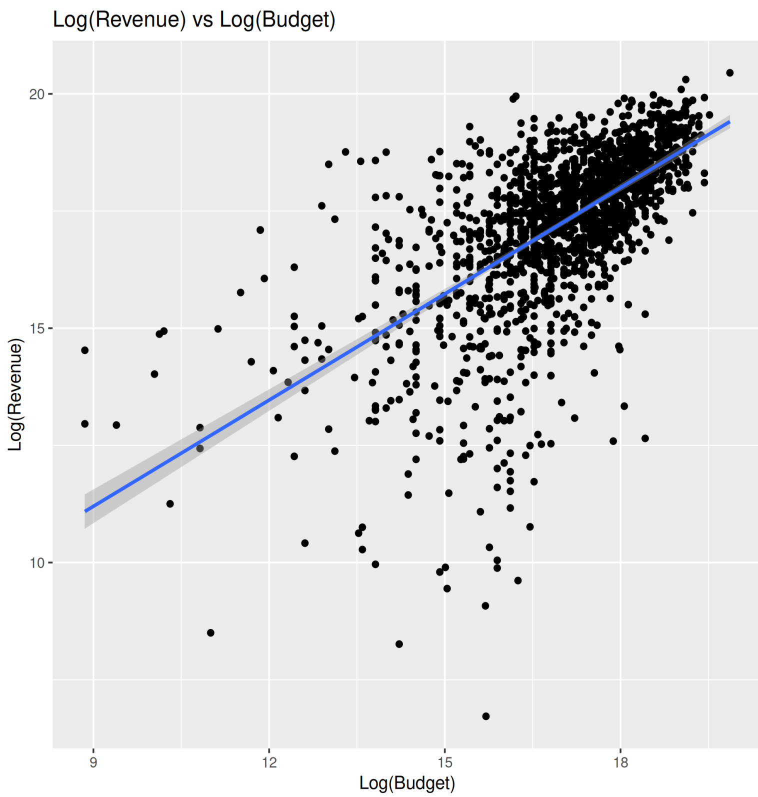

Figure 4: Scatterplot of log(Domestic Revenue) vs log(Movie Budget)

Methods and Results

Due to the relationship between movie budget and movie revenue being relatively linear, we will use a linear regression for analysis. Below, we split the data and summarize the number of observations in each training and test split. We also look at the mean values for the log-transformed budget and revenue columns in each training and test split.

Table 7: Mean log-budget and log-domestic revenue for the training set

train_log_budget_mean	train_log_domgross_mean
16.98953	17.24123

Table 8: Mean log-budget and log-domestic revenue for the test set

test_log_budget_mean	test_log_domgross_mean
16.9104	17.16077

Below we fit a simple linear regression with the log-transformed versions of movie budget and movie revenue to account for data skew. The predictor (log-transformed movie budget) is statistically significant.

```
Call:
lm(formula = log_domgross ~ log_budget, data = train)

Residuals:
    Min      1Q  Median      3Q     Max 
-9.5448 -0.5775  0.1807  0.7754  4.3182 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  4.33845    0.48805   8.889   <2e-16 ***
log_budget   0.75945    0.02863  26.524   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.391 on 1241 degrees of freedom
Multiple R-squared:  0.3618,    Adjusted R-squared:  0.3613 
F-statistic: 703.5 on 1 and 1241 DF,  p-value: < 2.2e-16
```

Using the fitted model above, we predict on the test set.

Table 9: Predicted log domestic revenue with the corresponding log-budget and actual log-domestic revenue for test set

log_domgross_preds	log_budget	log_domgross
17.72169	17.62217	16.41186
17.27529	17.03439	17.43463
17.41376	17.21671	17.37845
18.40050	18.51599	17.93840
16.98240	16.64872	17.80893
16.05300	15.42495	18.24139

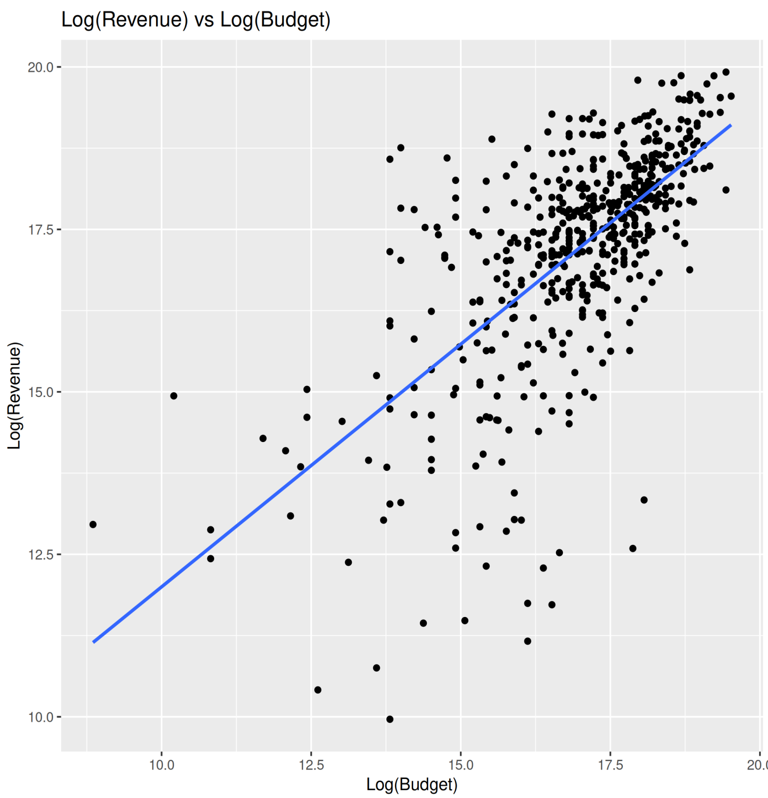

Figure 5: Scatterplot of log(movie budget) vs log(domestic revenue) for the test set

Table 10: RMSE of the linear regression model

.metric	.estimator	.estimate
rmse	standard	1.266987

The RMSE is 1.267. This means on average, the model's predicted log revenue differs from the actual log revenue by about 1.267. On average, the predicted revenue can be roughly 3.6 times higher or lower than the actual revenue.

K-Nearest Neighbours Regression model

Below, we also try fitting a K-nearest neighbours (KNN) regression model using cross-validation to compare whether RMSE might be improved compared to the linear model. First, we create the recipe and set log-transformed revenue as our dependent variable and log-transformed budget as our independent predictor variable, while scaling and centering the predictor.

Then, we set our model parameters to use KNN for regression, split the data into 5 folds for cross validation, and create a workflow with our model and recipe.

Table 11: Cross-validated RMSE for different K values in KNN regression (first 6 rows).

neighbors	.metric	.estimator	mean	n	std_err	.config
1	rmse	standard	1.808973	10	0.0594194	pre0_mod01_post0
4	rmse	standard	1.503846	10	0.0373065	pre0_mod02_post0
7	rmse	standard	1.429560	10	0.0404014	pre0_mod03_post0
10	rmse	standard	1.401511	10	0.0367946	pre0_mod04_post0
13	rmse	standard	1.394200	10	0.0384592	pre0_mod05_post0
16	rmse	standard	1.392741	10	0.0426348	pre0_mod06_post0

In Table 11, we test every 3rd number of neighbours between 1 and 200, set using tibble(), then filtered for the metric root mean standard error (RMSE) to evaluate the fit.

Table 12: The minimum RMSE.

neighbors	.metric	.estimator	mean	n	std_err	.config
91	rmse	standard	1.37291	10	0.0475376	pre0_mod31_post0

In Table 12, we filter the results to get the K value with the lowest associated RMSE.

Table 13: Cross-validated RMSE for KNN regression model at previously found K-value.

.metric	.estimator	mean	n	std_err	.config
rmse	standard	1.37291	10	0.0475376	pre0_mod0_post0

In Table 13, we find the cross-validated RMSE for the model at K = 91. For the training set, the average RMSE across the 10 folds is 1.37 log units with a standard error of 0.05.

Table 14: RMSE of the KNN regression model

.metric	.estimator	.estimate
rmse	standard	1.281567

From Table 14, the test RMSPE of 1.28 means that, on average, the KNN model's predicted log revenue differs from the actual log revenue by about 1.28. On average, the predicted revenue can be roughly 3.6 times higher or lower than the actual revenue.

So, both KNN and linear regression have similar predictive accuracy, with linear regression being slightly better.

Discussion

Our analysis suggests that movie budget is moderately associated with domestic box office revenue. The exploratory analysis showed a positive relationship between the two variables, supported by a correlation coefficient of approximately 0.63, indicating a moderate positive association. As the raw data is heavily right-skewed, applying a logarithmic transformation to both variables helped reduce the skewness and produced a more linear relationship suitable for modelling.

The linear regression model fitted on the log-transformed variables indicated that log-budget is a statistically significant predictor of log-revenue. This suggests that increases in movie budget tend to correspond with increases in domestic box office revenue on average. When evaluated on the test set, the model produced an RMSPE of approximately 1.27 log units, meaning predicted revenue may differ from the true revenue by roughly a factor of 3.55 on average.

To evaluate whether a more flexible model could improve prediction accuracy, we also trained a KNN model with cross-validation to tune the number of neighbours. The optimal value with K = 91, producing a test RMSPE of approximately 1.37, very similar to the linear regression model. Since KNN did not significantly improve predictive performance, this suggests that the relationship between log-budget and log-revenue is reasonably captured by the simpler linear model. Consistent with previous research on film economics, our results suggest that while

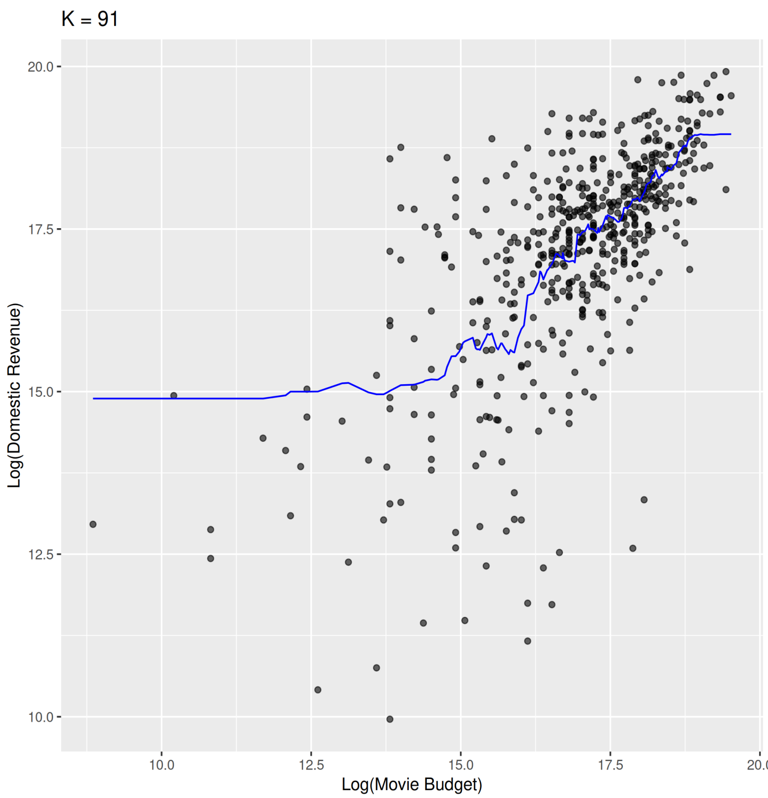

Figure 6: Predicted values of log(movie revenue) (blue line) for the final K-NN regression model

budget is associated with higher revenue, it alone cannot fully explain the variability in box office outcomes (Prag and Casavant 1994).

Despite the observed relationship, a couple of limitations exist in this analysis. First, the models only consider budget as a single predictor, when movie revenue is likely influenced by many additional factors such as marketing, release timing, critical reception, and star power. Second, the dataset only includes domestic revenue, where particular films may have performed differently internationally.

Future work could improve predictive performance by incorporating additional variables such as genre, production studio, critical reception, and by analyzing worldwide revenue rather than only domestic. More advanced modeling techniques such as random forests or gradient boosting models may also capture nonlinear relationships between features and revenue.

Overall, our findings suggest that movie budget does have a moderate predictive relationship with domestic revenue, but it is not a complete predictor of financial success.

Citations

Basuroy, Suman, Subimal Chatterjee, and S. Abraham Ravid. 2003. "How Critical Are Critical Reviews? The Box Office Effects of Film Critics, Star Power, and Budgets." *Journal of Marketing* 67 (4): 103–17. https://doi.org/10.1509/jmkg.67.4.103.18692.

De Vany, Arthur, and W. David Walls. 1999. "Uncertainty in the Movie Industry: Does Star Power Reduce the Terror of the Box Office?" *Journal of Cultural Economics* 23 (4): 285–318.

FiveThirtyEight. 2014. *Bechdel Test Movie List*. https://github.com/fivethirtyeight/data/tree/master/bechdel.

Prag, Jay, and James Casavant. 1994. "An Empirical Study of the Determinants of Revenues and Marketing Expenditures in the Motion Picture Industry." *Journal of Cultural Economics* 18 (3): 217–35.